

Analise e Desenvolvimento de Sistemas

Práticas DevOps

```
1287     $return[$day->shot_date] = $day;
1288 }
1289
1290 return $return;
1291 }
1292
1293 static function day_images_list($day, $studio) {
1294     global $global_studio_list;
1295     if(!in_array($studio, $global_studio_list)) die("error studio");
1296     $date = mysql::escape($day);
1297     if(mysql::count("image_date", "shot_date = '$date'") < 1) die('date not found');
1298     $studio = intval($studio);
1299
1300     $return = array();
1301
1302     $result = mysql::query("SELECT image.id as image_id, image, image_date WHERE image_date.id=image.day_id AND image_date.shot_date = '$date' AND image.enabled=1 AND image.studio = '$studio'");
1303     while($img = mysql::fetch($result)) {
1304         $img->copyright = metadata::get_copyright($img->image_id);
1305         $img->models = metadata::get_models($img->image_id);
1306         $return[$img->image_id] = $img;
1307     }
1308 }
```

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Explorando a Ferramenta GitHub

Módulo 3

- Versionamento de Código
- Pipelines no GitHub



Versionamento de Código

Unidade 3.1



O GitHub



- Criado em 2008
- Microsoft adquiriu em 2018 o GitHub por US\$ 7,5 bilhões
- Possui 36 milhões de usuários
- Possui 100 milhões de projetos
- Possui licença gratuita
- Possui Códigos abertos (públicos) e códigos fechados (privados)



O GitHub

- Armazena, Gere e Versiona inclusive arquivos comuns
- Colaboração e Comunicação
- Documentação e Visibilidade (Wikis, Views, Markdown, etc)
- Páginas de empresas e desenvolvedores (Redes Sociais)
- Integração com IA Generativa para criar código (CoPilot)
- Actions -> Para CI e CD
- Segurança: Code Scanning



Features of GitHub



- **GitHub Codespaces**
- **GitHub Advanced Code Search**
- **GitHub Actions**
- **GitHub Pages**
- **GitHub CoPilot**
- **GitHub Marketplace**
- **GitHub Pull Requests**
- **GitHub Discussions**
- **GitHub CLI**
- **Security Alerts**
- **Native Mobile Apps**

O GitHub



- Projetos famosos hospedados no GitHub:
 - <https://github.com/WordPress>
 - <https://github.com/mysql>
 - <https://github.com/grafana>
- Pessoas Famosas:
 - <https://github.com/addyosmani>
 - <https://github.com/tonsky>
 - <https://github.com/DanWahlin>



O GitHub

- Documentação Oficial GitHub

<https://docs.github.com/pt>



O GitHub – Cli e Desktop

<https://docs.github.com/pt/github-cli>

<https://docs.github.com/pt/desktop>



O GitHub – Clonar Repositório Público

Angular-HelloWorld Public Watch 2

main 1 Branch 0 Tags

Go to file t Add file <> Code

Local Codespaces

Clone ?

HTTPS SSH **GitHub CLI**

<https://github.com/DanWahlin/Angular-HelloWorld>

Clone using the web URL.

Open with GitHub Desktop

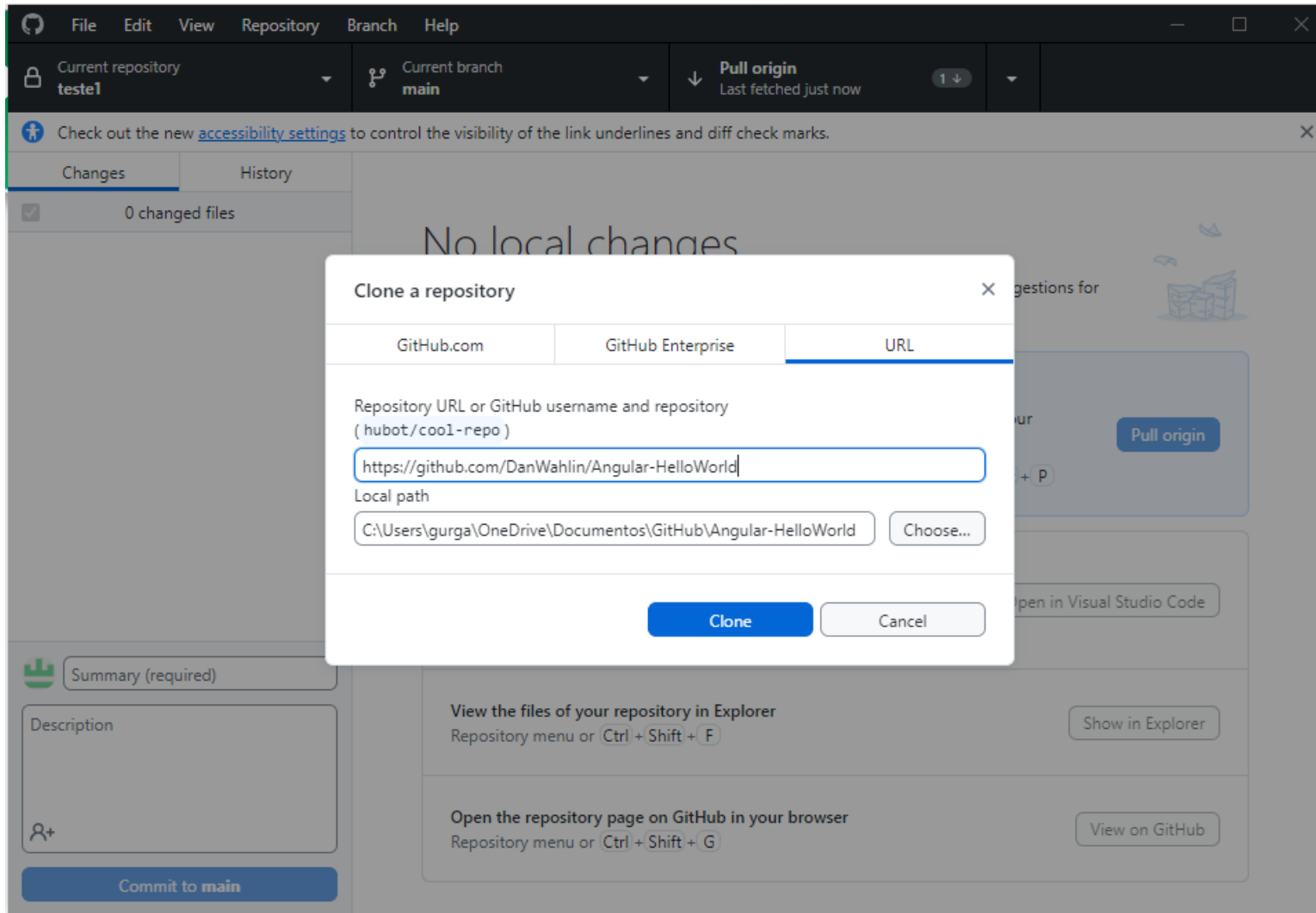
Download ZIP

Dan Wahlin	Update index.html
.vscode	Initial commit
src	Update index.html
.editorconfig	Initial commit
.gitignore	Initial commit
README.md	Initial commit
angular.json	Initial commit
package-lock.json	Initial commit
package.json	Initial commit
tsconfig.app.json	Initial commit
tsconfig.json	Initial commit
tsconfig.spec.json	Initial commit

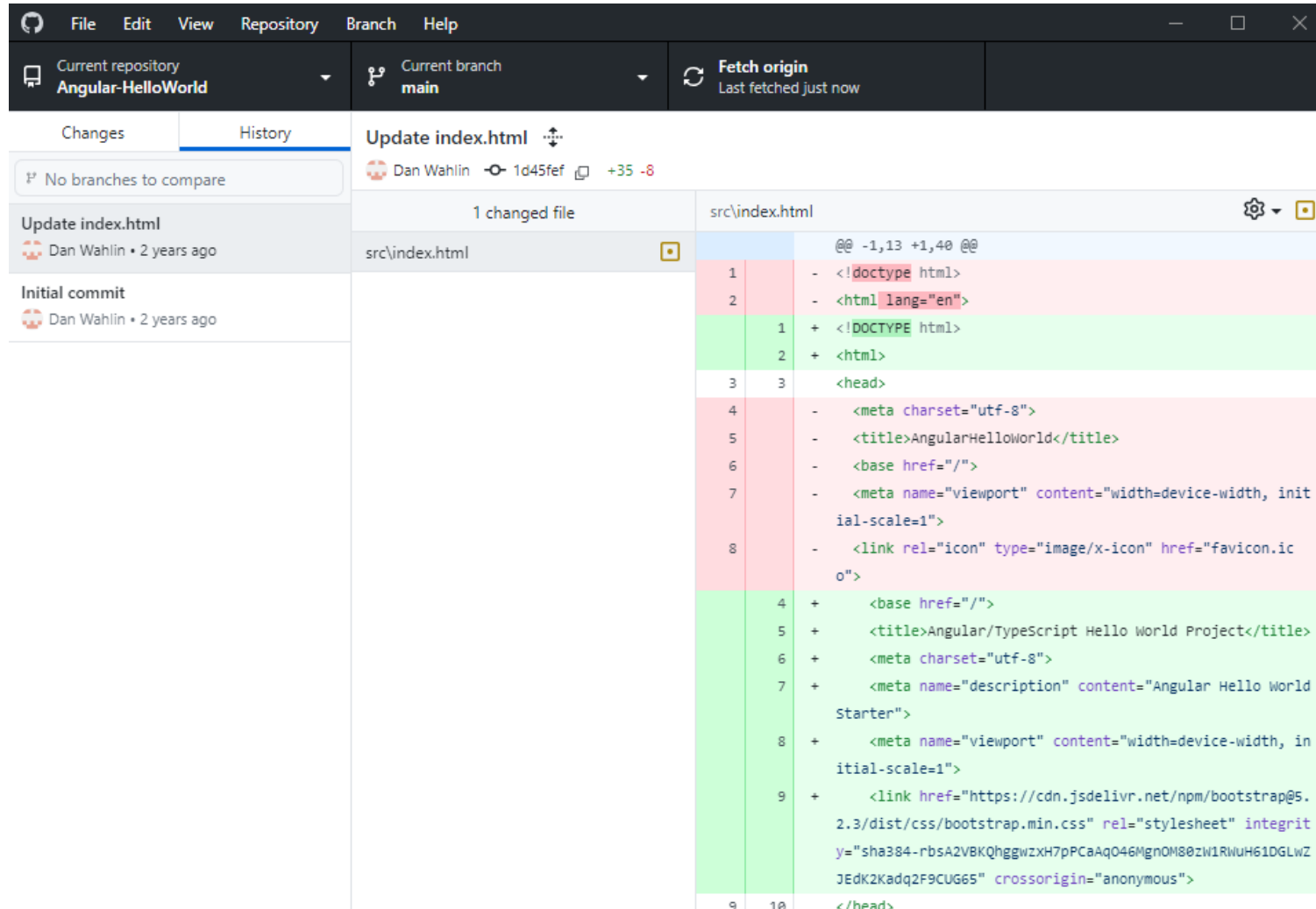
2 years ago



O GitHub – Clonar Repositório Público



O GitHub – Clonar Repositório Público



The screenshot displays the GitHub Desktop application window. The top bar shows the current repository as 'Angular-HelloWorld', the current branch as 'main', and a 'Fetch origin' button. The left sidebar has tabs for 'Changes' and 'History'. The 'History' tab is active, showing a list of commits. The first commit, 'Update index.html' by Dan Wahlin, is selected. The main area shows the diff for 'src\index.html', indicating 1 changed file. The diff view shows the changes between the previous commit and the current one. The file 'src\index.html' is shown with a diff view. The diff view shows the changes between the previous commit and the current one. The file 'src\index.html' is shown with a diff view. The diff view shows the changes between the previous commit and the current one.

Current repository: Angular-HelloWorld

Current branch: main

Fetch origin: Last fetched just now

Changes: No branches to compare

History:

- Update index.html (Dan Wahlin • 2 years ago)
- Initial commit (Dan Wahlin • 2 years ago)

Update index.html

1 changed file

src\index.html

@@ -1,13 +1,40 @@

```
1 - <!doctype html>
2 - <html lang="en">
3
4 + <!DOCTYPE html>
5 + <html>
6
7   <head>
8     - <meta charset="utf-8">
9     - <title>AngularHelloWorld</title>
10    - <base href="/">
11    - <meta name="viewport" content="width=device-width, initial-scale=1">
12    - <link rel="icon" type="image/x-icon" href="favicon.ico">
13
14 +   <base href="/">
15 +   <title>Angular/TypeScript Hello World Project</title>
16 +   <meta charset="utf-8">
17 +   <meta name="description" content="Angular Hello World Starter">
18 +   <meta name="viewport" content="width=device-width, initial-scale=1">
19 +   <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.2.3/dist/css/bootstrap.min.css" rel="stylesheet" integrity="sha384-rbsA2VBKQhggwzxH7pPCaAqO46MgnOM80zW1RWuH61DGLWZJEdK2Kadq2F9CUG65" crossorigin="anonymous">
20
21 </head>
```

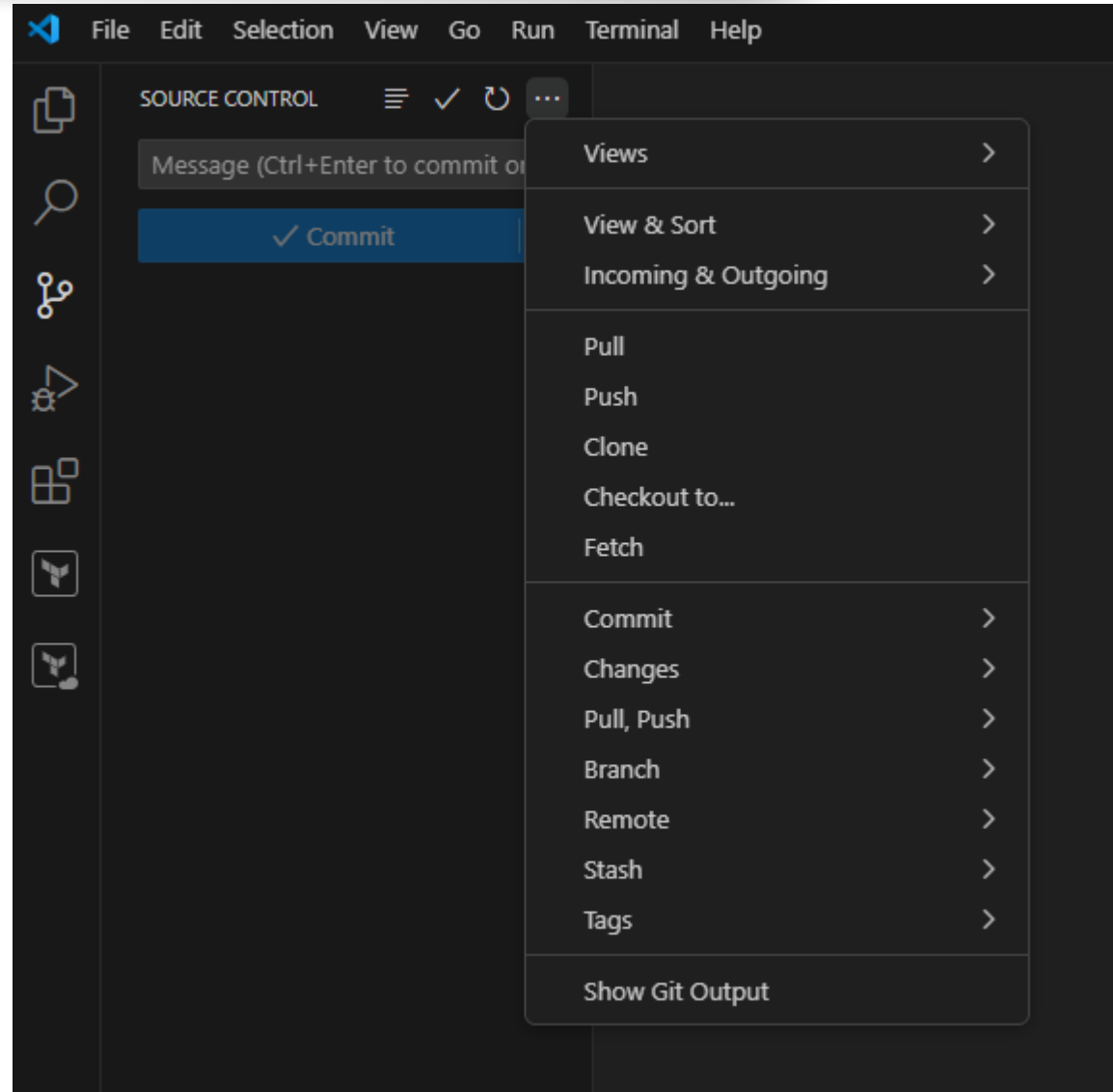


O GitHub – Passo a Passo

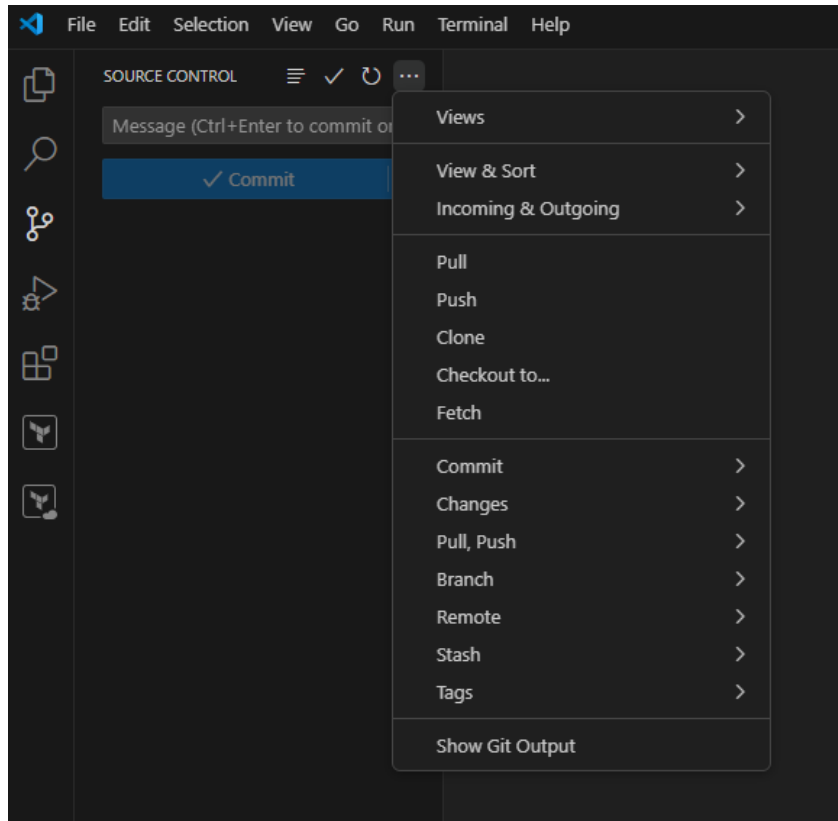
- Criar uma Conta
- Criar um Repositório
- Readme e a Linguagem Markdown
<https://www.markdownguide.org/basic-syntax/>



O GitHub – No VsCode



O GitHub – No VsCode



- **Clone:** Copia o repositório completo para o computador local
- **Commit:** Salva as alterações até o momento no repositório local. Permite voltar a esse ponto (antes do salvamento) caso necessário
- **Push:** Envia as alterações para a nuvem
- **Pull:** Baixa as atualizações de um repositório



O GitHub – No VsCode

- Executando Commit, Push e Pull no GitHub Usando VSCode



O GitHub – No VsCode

- **Resolvendo Conflitos no GitHub com VsCode**



O GitHub – Branches

- Criar um Branch
- Usar um Branch



Pipelines no GitHub

Unidade 3.2



Exemplos de Pipelines

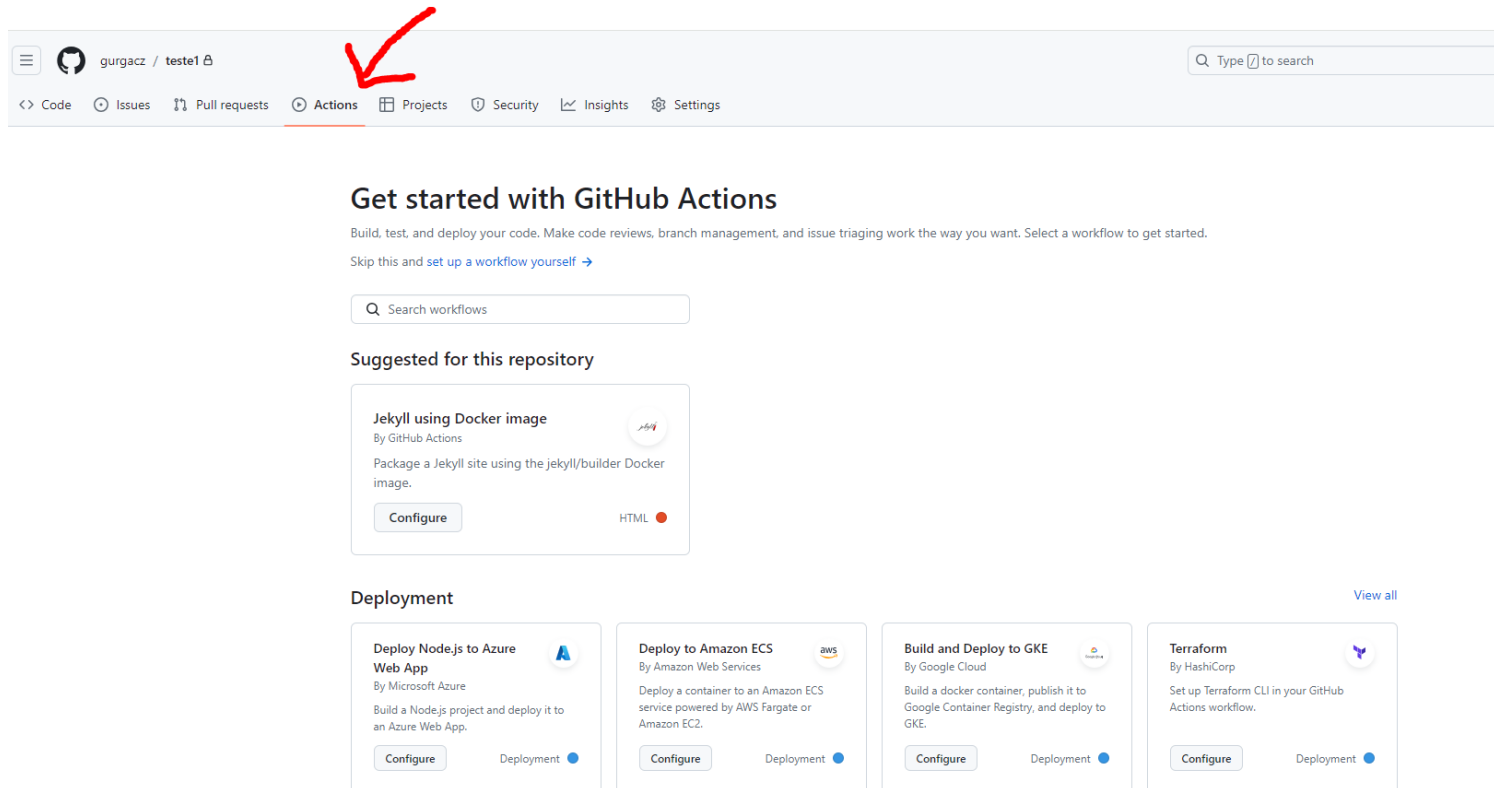


GitHub Actions

- Realizar o Build ou compilação do código para confirmar que não possui erros de escrita de código;
- Realizar uma bateria de testes unitários para identificar se não possui erros funcionais no sistema ;
- Realizar uma análise de padrões e boas práticas, tais como: nome de variáveis, quantidade de comentários, uso indevido de operações ou comandos;
- Realizar a execução do deploy do sistema em um ambiente qualquer;
- Enviar e-mails e alertas de operações;
- Solicitar aprovação de ações realizadas.



GitHub Actions



The screenshot shows the GitHub interface for the repository `gurgacz / teste1`. The **Actions** tab is highlighted with a red arrow. Below the navigation bar, the heading **Get started with GitHub Actions** is followed by a brief description and a link to [set up a workflow yourself](#). A search bar for workflows is present. Under the **Suggested for this repository** section, there is a card for **Jekyll using Docker image** by GitHub Actions, which includes a **Configure** button and a status indicator for HTML. The **Deployment** section features four cards: **Deploy Node.js to Azure Web App** (by Microsoft Azure), **Deploy to Amazon ECS** (by Amazon Web Services), **Build and Deploy to GKE** (by Google Cloud), and **Terraform** (by HashiCorp). Each card contains a description, a **Configure** button, and a status indicator for Deployment.

gurgacz / teste1

<> Code Issues Pull requests **Actions** Projects Security Insights Settings

Get started with GitHub Actions

Build, test, and deploy your code. Make code reviews, branch management, and issue triaging work the way you want. Select a workflow to get started.

Skip this and [set up a workflow yourself](#) →

Search workflows

Suggested for this repository

Jekyll using Docker image

By GitHub Actions

Package a Jekyll site using the jekyll/builder Docker image.

[Configure](#)

HTML ●

Deployment

[View all](#)

Deploy Node.js to Azure Web App

By Microsoft Azure

Build a Node.js project and deploy it to an Azure Web App.

[Configure](#)

Deployment ●

Deploy to Amazon ECS

By Amazon Web Services

Deploy a container to an Amazon ECS service powered by AWS Fargate or Amazon EC2.

[Configure](#)

Deployment ●

Build and Deploy to GKE

By Google Cloud

Build a docker container, publish it to Google Container Registry, and deploy to GKE.

[Configure](#)

Deployment ●

Terraform

By HashiCorp

Set up Terraform CLI in your GitHub Actions workflow.

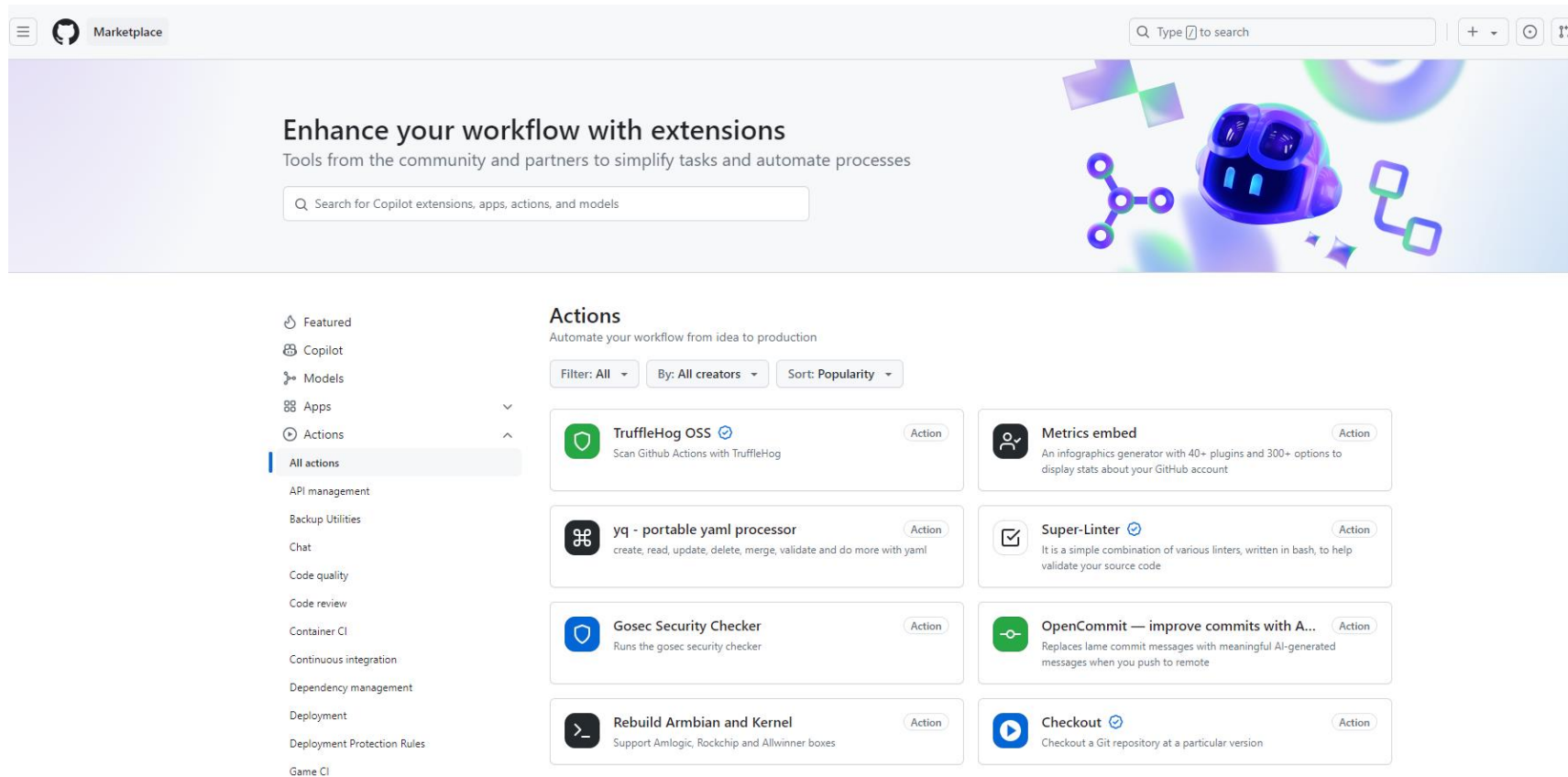
[Configure](#)

Deployment ●



GitHub Actions - Marketplace

<https://github.com/marketplace?type=actions>



The screenshot shows the GitHub Marketplace interface for Actions. At the top, there's a header with the GitHub logo and a search bar. Below the header, a banner reads "Enhance your workflow with extensions" with a subtext "Tools from the community and partners to simplify tasks and automate processes". A search bar is also present in the banner area. On the left, a sidebar lists various categories: Featured, Copilot, Models, Apps, Actions (selected), All actions, API management, Backup Utilities, Chat, Code quality, Code review, Container CI, Continuous integration, Dependency management, Deployment, Deployment Protection Rules, and Game CI. The main content area is titled "Actions" with the subtitle "Automate your workflow from idea to production". It includes filters for "Filter: All", "By: All creators", and "Sort: Popularity". Below the filters, there are six action cards displayed in a grid:

- TruffleHog OSS**: Scan GitHub Actions with TruffleHog. (Action)
- Metrics embed**: An infographics generator with 40+ plugins and 300+ options to display stats about your GitHub account. (Action)
- yq - portable yaml processor**: create, read, update, delete, merge, validate and do more with yaml. (Action)
- Super-Linter**: It is a simple combination of various linters, written in bash, to help validate your source code. (Action)
- Gosec Security Checker**: Runs the gosec security checker. (Action)
- OpenCommit — improve commits with A...**: Replaces lame commit messages with meaningful AI-generated messages when you push to remote. (Action)
- Rebuild Armbian and Kernel**: Support Amlogic, Rockchip and Allwinner boxes. (Action)
- Checkout**: Checkout a Git repository at a particular version. (Action)



GitHub Actions - YAML

- GitHub Actions usa a sintaxe YAML para escrever os Pipelines
- Cada pipeline é armazenado como um arquivo com extensão yml e deve ficar salvo no diretório chamado `.github/workflows`
- O cabeçalho base de qualquer pipeline escrito em YAML no GitHub é:

`name: nome_do_pipeline`

`on: [xxxxx]`

`jobs:`



GitHub Actions – Exemplo de um Pipeline em YAML

```
name: GitHub Pipeline de Exemplo
run-name: Teste de GitHub Actions
on: [push]
jobs:
  Explore-GitHub-Actions:
    runs-on: ubuntu-latest
    steps:
      - run: echo "Job iniciado automaticamente"
      - run: echo "Job executando no ${ runner.os } server hospedado pelo GitHub!"
      - run: echo "Branch ${ github.ref } e repositório ${ github.repository }."
      - name: Copiando código para o ambiente do GitHub Actions
        uses: actions/checkout@v4
      - run: echo "O repositório ${ github.repository } foi clonado no servidor do Github."
      - name: Listando arquivos copiados para o ambiente do GitHub Actions
        run: |
          ls ${ github.workspace }
      - run: echo "Situação do Job ${ job.status }."
      - run: echo "Pipeline Encerrado"
```



GitHub Actions – Exemplo de um Pipeline em YAML

```
name: GitHub Pipeline de Exemplo
run-name: Teste de GitHub Actions
on: [push]
jobs:
  Explore-GitHub-Actions:
    runs-on: ubuntu-latest
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      - run: echo "Job iniciado automaticamente"
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      - run: echo "Pipeline Encerrado"
```

Nome do Pipeline

Quando será executado.

Pull Request
Push
Commit
Create

Onde o pipeline será executado



GitHub Actions – Exemplo de um Pipeline em YAML

- Criar um exemplo do Pipeline com o código anterior
- Mostrar o resultado após o Push

```
name: GitHub Pipeline de Exemplo
run-name: Teste de GitHub Actions
on: [push]
jobs:
  Explore-GitHub-Actions:
    runs-on: ubuntu-latest
    steps:
      - run: echo "Job iniciado automaticamente"
      - run: echo "Job executando no ${ runner.os } server hospedado pelo GitHub!"
      - run: echo "Branch ${ github.ref } e repositório ${ github.repository }."
      - name: Copiando código para o ambiente do GitHub Actions
        uses: actions/checkout@v4
      - run: echo "O repositório ${ github.repository } foi clonado no servidor do Github."
      - name: Listando arquivos copiados para o ambiente do GitHub Actions
        run: |
          ls ${ github.workspace }
      - run: echo "Situação do Job ${ job.status }."
      - run: echo "Pipeline Encerrado"
```



GitHub Actions – Exemplo de um Pipeline Real

```
name: Pipeline Teste
on
  pull_request:
    branches:
      - main
jobs:
  Tarefa1:
    runs-on: ubuntu-latest
    steps:
      - name: Passo1-Puxar última versão do código
        uses: actions/checkout@v2
      - name: Passo2-Instalar Node.JS
        uses: actions/setup-node@v1
        with:
          node-version: "12.x"
      - name: Passo3-Instalar Dependencias
        run: npm ci
      - name: Passo4-Rodar os testes
        run: npm run test:ci
```



Dúvidas???

